

DEVELOPMENTAL BIOLOGY

A Notch signal required for a morphological novelty in *Drosophila* has antecedent functions in genital disc eversion

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The origin of morphological novelties has long fascinated biologists. Signaling pathways play important roles in the formation of novelties; however, the history of how they become integrated into new developmental programs remains unclear. Here, we investigated the evolution of the posterior lobe, a novel structure in the male genitalia of *Drosophila melanogaster*. We demonstrated that a Notch signaling center is required for the formation of this novelty and identified enhancers of the ligand *Delta*, which allowed us to track the evolutionary history of this signaling center. Unexpectedly, we found that the posterior lobe signaling center emerged from a preexisting role in genital disc eversion. We provide a likely mechanism by which *Delta* contributes to genital eversion through a network of apical extracellular matrix, which also became integrated into the posterior lobe program. This work demonstrates that novelties may be formed in the context of already complex developmental processes, by appending new roles to preexisting signals.

INTRODUCTION

The evolutionary origin of new morphological structures (“morphological novelties”) remains an enigmatic process that has captivated the imagination of biologists for centuries. Despite astounding anatomical diversity in the animal kingdom, genes that govern the formation of novelties are often conserved even between distantly related taxa (1–3) and recurrently used in multiple tissues over the course of an organism’s development (4, 5). Specifically, a core set of signaling pathways is frequently reused throughout development to regulate the formation of different tissues (6, 7). Though elegant work on complex morphological novelties such as the turtle’s shell (8, 9), butterfly eyespots (10, 11), the bat’s wing (12–14), and birds’ feathers (15) has implicated a prominent role of signaling pathway redeployment in their origination, we currently lack a molecular picture of how these roles first appeared and were shaped into complex genetic programs. In particular, we need detailed histories of how important developmental signals became expressed to pattern these structures and how downstream cellular responses first emerged.

To study the origination of a novel structure, two factors have hindered progress toward developing a satisfying molecular history of how novelties were assembled. First, many novel morphologies of interest have evolved over long evolutionary times, such that the ancestral context from which the novelty first emerged cannot be traced with any confidence (9, 16). Some developmental evolutionary studies of novelty have focused on the development of single organisms without comparing them to outgroup species that lack the structure (9, 10, 17, 18). This is an understandable approach for macroevolutionary changes that arose in the distant past, as conserved landmarks

to reference homologous tissues are often absent. However, focusing on ancient novelties may overlook multiple layers of change that erased informative intermediate steps along their evolutionary trajectories. Thus, evaluating novelties across a wide range of timescales may offer multiple unique insights into their beginnings. A second barrier is that most systems lack the genetic and developmental tools necessary to assess gene regulatory changes and the function of genes in the relevant tissue (8, 19, 20). Both of these barriers have necessitated the study of more recently evolved morphological novelties that exist in genetically tractable model organisms (21–30). Doing so allows us to assess and pinpoint changes within the gene regulatory networks (GRNs) underlying these novelties.

GRNs control development by integrating spatial and temporal information from transcription factors and signaling pathways to pattern the expression of target genes (31). In the context of morphological novelty, the study of enhancers uniquely provides access to study mutations that may have contributed to the evolution of these traits through comparative reporter assays (22, 27). By evaluating the activities of orthologous regulatory regions from two or more species in a common genetic background, one can attribute activity differences to the tested regulatory elements. Furthermore, reporter assays can detect co-option and pleiotropy (22, 32, 33). When a program downstream of a signaling pathway is deployed to a new context, the responsible enhancer will be pleiotropic for both ancestral and novel tissues (22). Last, enhancer elements are particularly useful for tracing cellular lineages through complex morphogenetic movements that may be difficult to disentangle.

The genitalia of the model organism *Drosophila melanogaster* offers a system in which a relatively recent morphological novelty can be examined in a genetically tractable system. Genital traits are noteworthy for their rapid evolution (34) and are often the distinguishing characteristic between closely related species (35). Notably, the posterior lobe is a morphological novelty in the male genitalia of *D. melanogaster* (35, 36) (Fig. 1, A and B). This cuticular outgrowth is unique to the *melanogaster* clade and is necessary for copulation (35, 37, 38). The posterior lobe projects from an ancestral genital tissue called the lateral plate [also known as the epandrial ventral

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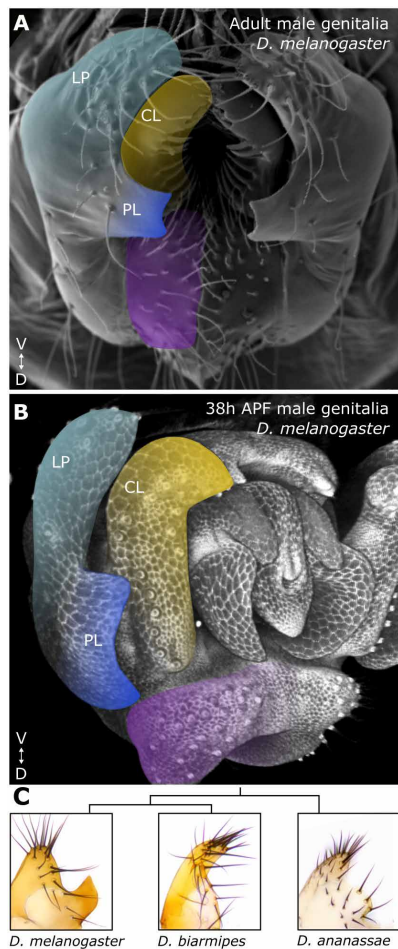


Fig. 1. Male genitalia of *D. melanogaster*. *D. melanogaster* adult (A) and developing pupal [38 hours (h) after pupal formation (APF)] (B) male genitalia. The novel posterior lobe (royal blue) protrudes from an ancestral structure called the lateral plate (light blue), adjacent to another structure conserved between lobed and nonlobed species, the clasper (yellow). The analia is highlighted in purple. (C) Cladogram with bright-field images of dissected adult lateral plates of a lobed and two nonlobed species.

lobe (39)] and forms by locally increasing the apical height of lateral plate cells to generate the lobe (30). This structure evolved approximately 11.6 million years ago (40), and the genitalia of lobed and nonlobed species are composed of similar structures otherwise. This permits us to perform comparative analyses of gene expression and gene regulation during development between *D. melanogaster* and outgroup species that lack a lobe, such as *Drosophila ananassae* and *Drosophila biarmipes*, which serve as a developmentally accessible proxy for the ancestral ground state (Fig. 1C and fig. S1).

Here, we investigated how a signaling pathway became associated with the novel posterior lobe. We found that Notch signaling plays an important role for posterior lobe development. Specifically, the spatial expansion of the Notch ligand, Delta, in a zone adjacent to the posterior lobe is required for its development, and this expansion is unique to *D. melanogaster*. We dissected the regulatory elements involved in the deployment of Delta to the lobe-forming region and investigated its ancestral function in the development of genital structures. Unexpectedly, our analysis revealed that the Delta/Notch signaling center becomes active days before the posterior lobe forms,

serving a role in the development of conserved genital tissues. In particular, we have implicated an early-acting role for this signal in controlling genital disc eversion—a process by which the epithelium underlying these structures turns inside out. This work demonstrates that novelties may be formed in the context of ancestrally complex developmental programs, by adding new roles to preexisting signals to connect a new program to well-established ancestral ones.

RESULTS

Species-specific expression of *Delta* is essential for the development of the posterior lobe

In a screen of major signaling pathway ligands during posterior lobe development, we found that the Notch ligand, Delta, is expressed in multiple male genital structures, including a region adjacent to the developing posterior lobe at the base of the lateral plate and clasper [also known as the surstylus (39)] (Fig. 2A and fig. S2, A to C). This expression precedes posterior lobe development in the early pupal genitalia, as expected for a developmentally instructive signal

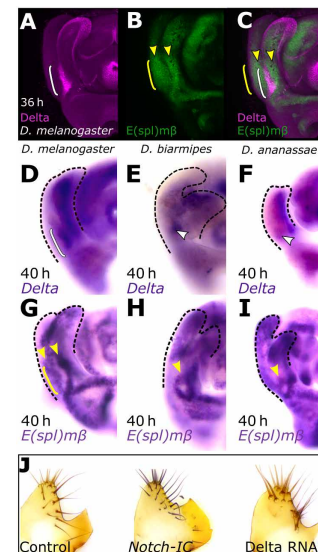


Fig. 2. The spatially expanded expression of *Delta* is necessary for posterior lobe development in *D. melanogaster*. (A) Immunofluorescence of Delta protein expression (magenta). Delta expression is present adjacent to the developing posterior lobe at 36 hours APF in *D. melanogaster* in a spatially expanded pattern (white bracket). (B and C) An *E(spl)mβ*-GFP reporter reveals Notch activity adjacent to Delta-expressing cells (yellow arrowheads) and the cells of the developing posterior lobe (yellow bracket), and acts as a readout of Notch signaling. (D to F) Comparison of *Delta* mRNA expression between *D. melanogaster* (D) and the nonlobed species *D. biarmipes* (E) and *D. ananassae* (F). The expression of *Delta* expands in *D. melanogaster* (bracket), but is localized to a smaller region in nonlobed species (arrowheads, F and G). (G to I) Comparison of a canonical Notch target, *E(spl)mβ*, mRNA expression between *D. melanogaster* (G), *D. biarmipes* (H), and *D. ananassae* (I). The cells of the posterior lobe are Notch responsive (yellow bracket). Two regions adjacent to either side of Delta-expressing cells respond to Notch signaling (yellow arrowheads) (G). In nonlobed species, Notch activity is limited to a small circular pattern at the base of the lateral plate and clasper (yellow arrowhead in I and J). Represented images of (A) to (I) display half of a pupal genitalia. (J) Knock-down of Delta reduces the size of the posterior lobe (right), and overactivation of the Notch pathway increases the size of the posterior lobe (middle) compared to the control (left). h, hours.

(fig. S2A). As the posterior lobe initiates its development at mid-pupal stages, the expression pattern of Delta spatially expands along the lateral plate and clasper boundary (fig. S2B). Once the posterior lobe forms, Delta's expression retracts dorsally toward the anal plate (fig. S2C). The early expression of Delta preceding posterior lobe development and its expansion corresponding to the developmental timing of the posterior lobe made it a strong candidate regulator of the posterior lobe GRN. To determine whether this pattern is unique to lobe-bearing species, we next examined the expression of Delta in *Drosophila* species that do not form a posterior lobe ("nonlobed" species). Immunofluorescent staining of Delta using a polyclonal antibody that is cross-reactive in multiple species, as well as in situ hybridization of *Delta* mRNA, revealed that Delta is expressed in a small area at the base of the claspers and lateral plates in the nonlobed genitalia of *D. biarmipes* and *D. ananassae* (Fig. 2, E and F, and fig. S2, D to F). This expression pattern is limited to a much smaller zone compared to *D. melanogaster* (Fig. 2, D to F, and figs. S2, D to F, and S3, G to I and J to L). These results suggested that the expansion of Delta is specific to the posterior lobe-forming species, *D. melanogaster*, and correlates with the timing of posterior lobe development.

We next assessed whether the expanded pattern of posterior lobe-associated Delta is required for lobe development. A Delta-directed short hairpin RNA was driven by a genital-specific GAL4 driver of *Pox neuro* (*Poxn*) (41), which is active in a broad pattern in the genitalia, including the base of the lateral plate and claspers where *Delta* is also expressed (fig. S2G) (22, 41). Comparisons of adult phenotypes revealed that reduction of *Delta* expression interferes with posterior lobe development, resulting in smaller and defective posterior lobes (Fig. 2J and fig. S2, H and L to P). Reduction of *Delta* expression using this driver results in a pattern that resembles Delta expression in nonlobed species, suggesting that some aspect of the spatially expanded pattern of Delta is necessary for posterior lobe development (fig. S2, E, F, and J). To further investigate the role of Delta-Notch in the development of the posterior lobe, we stimulated Notch pathway activity by expressing a constitutively active form of Notch (Notch intracellular domain) (42, 43) under the control of the aforementioned *Poxn* driver. These animals developed a larger posterior lobe compared to the controls (Fig. 2J and fig. S2H). These data showed that not only is the spatially expanded expression of *Delta* required for posterior lobe development, but that the size of the posterior lobe is sensitive to the amount of Notch signaling, suggesting that expansion of this pathway could have been an important evolutionary step in the origins of this structure.

To ascertain how the Notch pathway contributes to posterior lobe development, we assessed the spatial extent of its activation in the developing male genitalia. Delta is a transmembrane ligand of Notch, sending signals to adjacent cells (44, 45). This cell-cell signaling is noteworthy because the expression of Delta expands between the lateral plate and clasper, which is adjacent to the developing posterior lobe. Thus, we expect the cells of the posterior lobe to be Notch responsive, receiving signals from the adjacent Delta pattern. To identify an appropriate readout of Notch activity, we tested the expression of canonical Notch targets of the *Enhancer of split* [*E(spl)*] complex by in situ hybridization (42). Throughout the entire genitalia, we found regions of basic Helix-Loop-Helix repressor *E(spl)mβ* expression adjacent to regions of Delta expression, indicating that *E(spl)mβ* acts as an appropriate marker for Notch activity (Fig. 2G and figs. S3 and S4). Notably, in *D. melanogaster*, *E(spl)mβ* is expressed in two

patches of cells adjacent to each side of the expanded Delta pattern, including the posterior lobe developing from the lateral plate (Fig. 2G and figs. S3 and S4). To directly compare *E(spl)mβ* with Delta, we constructed an *E(spl)mβ*-GFP transcriptional reporter transgene containing the proximal 1.4 kilobase (kb) of sequence adjacent to its promoter, a region known to recapitulate *E(spl)mβ* in other imaginal tissues (46). We observed that Delta and *E(spl)mβ*-GFP are indeed active in mutually exclusive regions and that the developing lobe showed apparent Notch pathway activation (Fig. 2, B and C). To investigate Notch activity in nonlobed species, we tested the expression of *E(spl)mβ* in pupal genitalia of the nonlobed species *D. biarmipes* and *D. ananassae* by in situ hybridization. Similar to *D. melanogaster*, we found *E(spl)mβ* expression adjacent to Delta-expressing cells, but the expression of *E(spl)mβ* was limited to a small ring-like pattern at the base of the lateral plates and claspers. The region of this ring-like pattern of *E(spl)mβ* mRNA appeared to anti-correlate with Delta's spatially restricted expression patterns in these nonlobed species (Fig. 2, H and I, and figs. S3 and S4). These observations indicate that (i) the cells of the posterior lobe are indeed Notch responsive and that (2) Notch/Delta signaling has a conserved ancestral pattern of downstream pathway activity during the development of genital structures.

Partially redundant transcriptional enhancers regulate the lobe-associated expression of Delta

Considering the unique association of Delta's expanded expression with lobe development, we sought to discover how this novel deployment occurred. To determine whether changes in *Delta* expression were encoded by cis-regulatory evolution, we first identified the enhancer sequences regulating *Delta* specifically in the lobe-forming region. To this end, we carried out a screen of ~125 kb of DNA at the *Delta* locus for pupal genital enhancers (Fig. 3). We used transgenic reporters from the Janelia GAL4 collection (47), which spanned ~64 kb of noncoding DNA upstream and intronic sequences of *Delta*. In addition, we cloned ~95 kb of noncoding DNA downstream of *Delta* into transgenic GFP (green fluorescent protein) reporter constructs (Fig. 3A). Among these constructs, we identified two elements that recapitulate endogenous posterior lobe-associated expression of Delta (Fig. 3, A to C). Enhancer1 is located ~36 kb downstream of the *Delta* transcription start site (Fig. 3, A and B), and enhancer2 is ~60 kb downstream of the *Delta* coding transcription start site (Fig. 3, A and C).

Having located the posterior lobe enhancers of *Delta*, we next investigated their functions and necessity in posterior lobe development. CRISPR-Cas9-mediated homology-directed repair was used to delete each enhancer individually. For each deletion, the enhancer region was replaced with a Ds-Red marker driven by a 3X-P3 promoter, which permitted the detection of integration events by eye fluorescence. To our surprise, neither enhancer deletion (2.4 kb Δenhancer1 or 5.5 kb Δenhancer2) reduced the size of the posterior lobe (fig. S5A) or noticeably affected the expression of *Delta* (fig. S5B). Notably, the minimal DNA sequences sufficient to drive reporter activity for enhancer1 and enhancer2 were ~730 and ~900 bp, respectively (fig. S6), and our deletion assays targeted a larger region to ensure the removal of potential nearby regulatory sequences important for the activity of these enhancers endogenously. These results suggest that the two enhancers may act redundantly in regulating *Delta* near the posterior lobe.

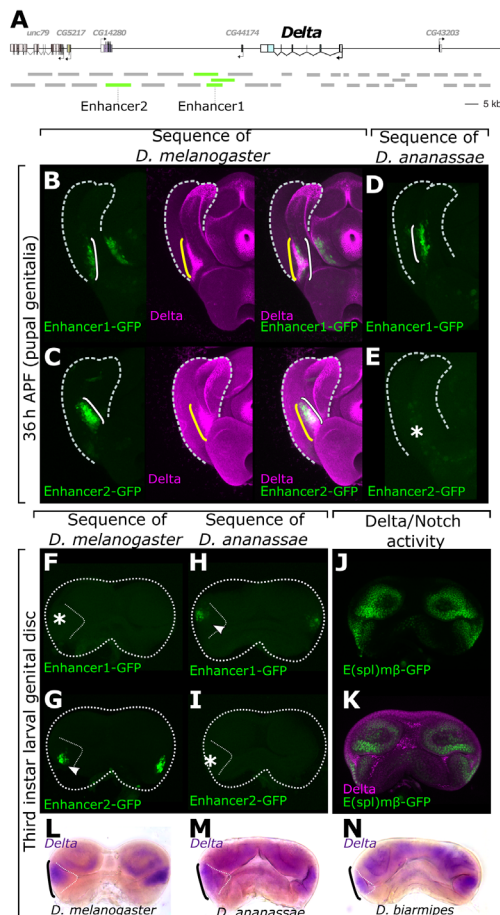


Fig. 3. Two posterior lobe-associated enhancers of *Delta* are temporally stratified and have partially redundant spatial activities in the pupal genitalia.

(A) Schematic of the *Delta* locus, with two enhancer regions displayed by green bars downstream of *Delta*. The smaller green bar of enhancer1 marks the overlap of two larger fragments. Gray bars indicate tested regulatory regions in the locus. (B and C) Activity of the posterior lobe-associated enhancers of *Delta* (green, white bracket) relative to the endogenous expression of *Delta* (magenta, yellow bracket). Enhancer1 overlaps the full expansion of *Delta* (B) whereas enhancer2 recapitulates the more ventral portion of the pattern (C). (D and E) Changes both in cis and trans contribute to the expanded expression of *Delta*. The orthologous region of enhancer1 from *D. ananassae* drives expression in an expanded pattern (bracket, D) similar to *D. melanogaster* (bracket, B) suggesting this element is functionally conserved. The orthologous segment of enhancer2 from *D. ananassae*, however, is not (or weakly) active in a *D. melanogaster* background (asterisks, E), suggesting changes to the element itself. (F and G) Enhancer2 (arrowhead, G), but not enhancer1 (asterisk, F), is active in the larval genital imaginal disc. (H and I) The orthologous sequence of enhancer1 but not enhancer2 of *D. ananassae* drives larval disc activity in a *D. melanogaster* background. (J and K) *Delta* immunofluorescence and *E(spl)mβ* reporter activity reveal Notch activity in the larval genital disc (J) and *Delta* expression that recapitulates reporter activity in the larval disc (K). (L to N) In situ hybridization of *Delta* mRNA (black brackets) confirms *Delta* is expressed in the larval genital disc of *D. melanogaster* (L), *D. ananassae* (M), and *D. biarmipes* (N). h, hours.

The *Delta*/Notch signaling center is active days before the posterior lobe forms

While the two enhancers of *Delta* may act redundantly and both overlap the endogenous expression of *Delta*, they exhibit slightly different spatial activities. In contrast to enhancer1, which fully recapitulates the endogenous expression pattern of *Delta* during posterior

lobe development, enhancer2 is active in cells located more ventrally, between the lateral plate and clasper (Fig. 3, B and C). Considering that expression patterns can change over developmental time, we were curious to know whether the posterior lobe-associated enhancers of *Delta* showed differences in their timing. Thus, we investigated the relative developmental timing of each enhancer's activation. To capture reporter activity more directly than GFP protein, which persists for hours after expression, we performed in situ hybridization to detect *GFP* transcripts in the transgenic reporter lines (48) (fig. S7A). This confirmed that the late activity of both enhancer1 and enhancer2 in the pupal genitalia represents active transcription rather than perdurance of GFP from an earlier stage. At earlier stages of pupal development, when the precursor of the posterior lobe has not yet differentiated, we detect stronger GFP mRNA signal from enhancer2 compared to enhancer1 (fig. S7A). Considering that the posterior lobe develops during mid-pupal stages, the earliest pupal stages we had previously assessed were at 24 hours APF (after pupal formation). The reasoning behind this was that at 24 hours APF, the lateral plate, which is the precursor of the posterior lobe, has not fully separated from the clasper. In addition, the developing genitalia of lobed and nonlobed species at this time point look quite similar, with the presumptive lateral plate and clasper still being fused together. Last, we were unable to detect obvious differences in *Delta* expression between lobed and nonlobed genitalia at 24 hours APF. However, the notable difference between enhancer1 and enhancer2 activity at 24 hours APF prompted us to assess even earlier pupal time points. To our surprise, enhancer2 was active throughout pupal development (Fig. 4A). To find the onset of enhancer2 activation, we tested its activity in a preceding stage of the *Drosophila* life cycle, the third instar larvae (L3), when the genital tissue is still an unevolved imaginal disc. Unexpectedly, enhancer2 drove GFP expression in the male genital primordium of the imaginal disc in a bilaterally symmetric pattern (Fig. 3G), but GFP was undetected for the enhancer1-GFP reporter (Fig. 3F). We confirmed that the larval disc activity of enhancer2 recapitulates the endogenous expression of *Delta* by in situ hybridization and immunofluorescence (Figs. 3, K and L, and 4D and fig. S8). The *E(spl)mβ*-GFP reporter was active in mutually exclusive regions with *Delta* in the larval genital disc (Fig. 3J), suggesting that Notch activity in response to the signaling center begins during prepupal development. The early imaginal disc activity of *Delta* and enhancer2 was a particularly unexpected finding, as this stage occurs 3 days before the transition to the pupal stage, and about $3\frac{1}{2}$ days before the initiation of posterior lobe development (49). Expectedly, this early activity is conserved between *D. melanogaster* and the nonlobed species *D. ananassae* and *D. biarmipes* (Fig. 3, L to N, and fig. S8), suggesting that an early ancestral role of *Delta* exists that has been expanded upon and modified in *D. melanogaster* to produce a pattern required for the formation of the novel posterior lobe.

Having identified an early role for enhancer2 in the larval imaginal disc, we reasoned that deletion of this enhancer would eliminate the early conserved activity of *Delta* and thus illuminate the ancestral role of this signaling center. However, as mentioned earlier, deletion of this enhancer did not affect *Delta* expression or the development of the posterior lobe. Furthermore, the early ancestral pattern of *Delta* was unaffected in the male genital primordium in Δ enhancer2 animals (fig. S5B). The lack of an obvious phenotype in this deletion line motivated us to further examine the *Delta* locus for

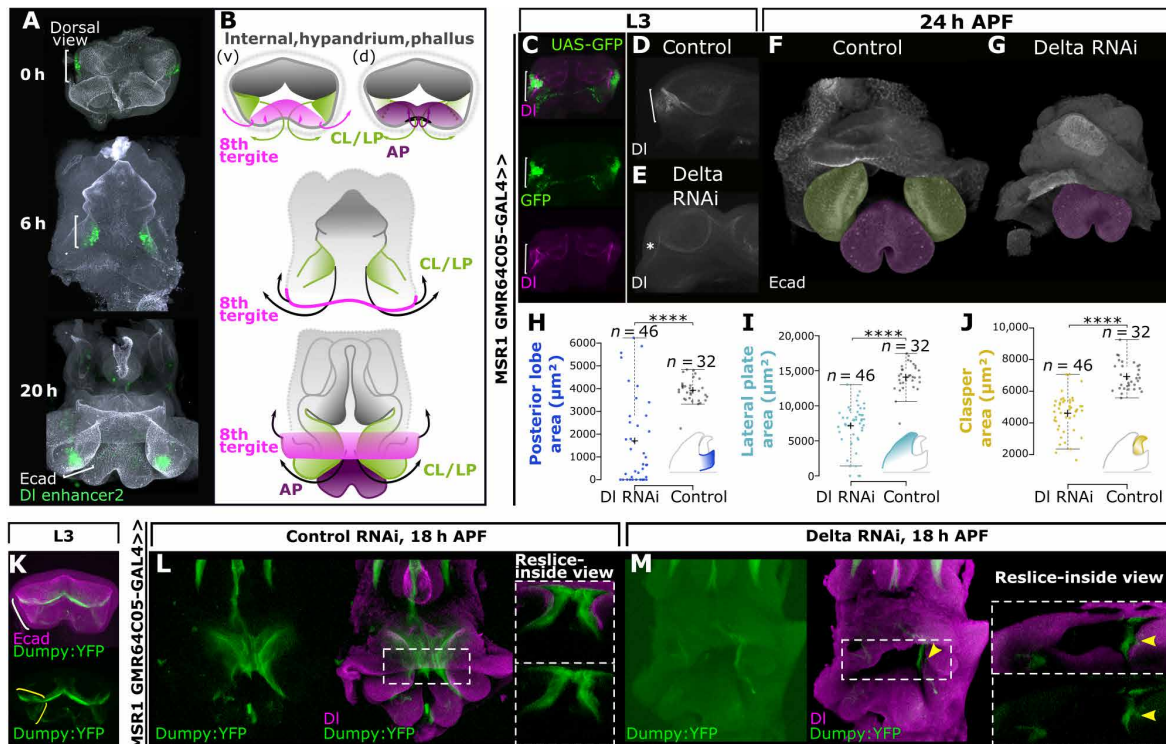


Fig. 4. The posterior lobe-associated signaling center has an early ancestral role involved in genital disc eversion. (A) The activity of enhancer2 tracks with the cells of the genital disc that undergo evagination. (B) Cartoon description of (A). The male genital primordium on the lateral sides of the L3 disc where Delta is active is denoted by green shading through development. The eighth tergite on the ventral side of the disc is colored pink. This tissue everts along with the future clasper/lateral plate (CL/LP) and eventually fuses dorsally. The anal plate is colored purple. Gray color marks tissue that will form the internal genitalia as well as the hypandrium, phallus, and branches. (C) Reporter activity of the MSR1-GMR64C05-GAL4 driver (green) in the L3 genital disc counterstained with Delta (magenta). (D and E) Reduction of Delta by RNAi in the L3 genital disc (E, asterisks) compared to control (D, bracket). (F and G) Early knockdown of Delta in the L3 genital disc causes defects in genital eversion (G) compared to a control (F). The anal plate is false colored in purple, and future CL/LP in the control is false colored in green. (H to J) Reduction of Delta early in the L3 genital disc causes severe defects in the development of the posterior lobe (H), lateral plate (I), and clasper (J). Asterisks denote significant differences (two-tailed Student's *t* test, *****P* < 0.0001). (K) Dumpy is deposited within the L3 genital disc and covers the epithelial folds of the lateral male genital primordium. The lateral fold is marked by a white bracket (top) and Dumpy deposition is marked by yellow brackets (bottom). (L and M) Early knockdown of Delta with an L3 genital disc driver reduces and disrupts the organization of Dumpy tethers [yellow arrowheads in (M)]. h, hours.

additional enhancers, focusing on the larval genital disc. Using the *Janelia* GAL4 collection (50), we identified seven additional upstream and intronic regions that drive larval disc activity (fig. S10). While the expression domain of some of these reporters is quite broad, they all overlap Delta's expression in the male genital primordium bilaterally on each side of the disc. Thus, it is likely that some of these enhancers work together or redundantly to regulate the early larval disc activity of Delta, suggesting that an early conserved role of Delta is controlled by a robust regulatory mechanism.

Changes within the *Delta* locus, as well as the trans-regulatory environment, contributed to the expansion of *Delta*

Given that enhancer1 and enhancer2 drive strong pupal activity resembling Delta's expression, we next examined the evolutionary history of these two enhancers. The spatial expansion of Delta in lobed species could be explained by two main mechanisms. First, mutations within the cis-regulatory elements may have led to Delta's deployment in a larger population of cells in lobed species. Second, changes to Delta-regulating transcription factors may have caused this spatial expansion. To understand the evolutionary history of

Delta's posterior lobe-associated expansion and test these two possibilities, we examined the ability of the orthologous DNA fragments from the nonlobed species *D. ananassae* and *D. biarmipes* to drive GFP in a common *D. melanogaster* trans-landscape. Each DNA fragment was cloned into a GFP reporter construct and inserted into the same landing site in the genome of *D. melanogaster*. We found that the orthologous sequence of enhancer1 from each species is able to drive GFP in a pattern resembling Delta's expanded expression (Fig. 3D and fig. S7E). This recapitulation suggests that enhancer1 is functionally conserved and that changes within the trans-regulatory landscape have contributed to the expansion of Delta. In contrast to enhancer1, the orthologous sequence of enhancer2 from nonlobed species did not activate GFP in a *D. melanogaster* background, suggesting evolutionary modifications to this element (Fig. 3E and fig. S4, B to D).

During larval development, enhancer1 sequences from nonlobed species drove strong larval disc activity (Fig. 3H and fig. S7E), unlike the *D. melanogaster* enhancer1. Enhancer2, on the other hand, drove little to no activity (Fig. 3I and fig. S7E). These results suggest that while the genital imaginal disc activity of Delta is ancestral, the functionally conserved enhancer1 has lost an ancient role

while the evolutionarily novel enhancer2 has taken over this conserved role. Collectively, comparative analysis of both enhancers indicated that a combination of cis- and trans-regulatory changes were responsible for the expansion of *Delta* to form the posterior lobe, and the two enhancers exhibit temporal shifts in the responsibility to drive the early conserved activity of *Delta*.

Delta plays an ancestral role in the eversion of the genital disc

We hypothesized that *Delta* is responsible for an ancestral function because of the following considerations: (i) *Delta* was expressed in the imaginal genital discs of both lobed and nonlobed species, (ii) *Delta*'s expression persisted in the developing pupal genitalia of nonlobed species, and (iii) Notch signaling was active in regions adjacent to *Delta*-expressing cells in the developing nonlobed genitalia. Furthermore, there is not a clear relationship between larval disc folds and pupal genital structures, and we were uncertain about which pupal structures the *Delta*-expressing lateral regions of the L3 primordium corresponded to. Thus, we sought to investigate the ancestral function of *Delta* to understand the origin of this signaling center and reasoned that tracing the lineage of *Delta*-expressing cells would provide insight into its ancestral role. We performed a developmental time course of enhancer2 activity from the genital imaginal disc through pupal development, staining with an antibody against E-cadherin to mark the apical surface of epithelial cells. The genital imaginal disc starts inside out, with the apical surfaces facing each other and the basal lamina surrounding the outside of the disc. As the disc develops, this tissue turns partially inside out like a pillowcase, with external structures emerging from the posterior edge to form the external anal and genital structures, such as the lateral plates and claspers (51). At 0 hours APF, enhancer2 is active in the male genital primordial folds on the lateral sides of the disc (Fig. 4, A and B). The folds in which enhancer2 is active correspond to regions that begin to evert outward from the opening at the posterior stalk at early pupal stages (Fig. 4, A and B). At around 20 hours APF, the cells marked by enhancer2 have partially everted, which form the precursor of the lateral plate and clasper (Fig. 4, A and B). These results demonstrated that *Delta*'s expression tracks with cells that undergo eversion to form the lateral plates and claspers, structures that are ancestral to the posterior lobe. Tracking the early larval disc expression of *Delta* with enhancer2 in *D. melanogaster* suggested that it is the same signaling center that persists through pupal development and later becomes precisely those *Delta*-positive cells adjacent to the posterior lobe as it forms.

To elucidate the ancestral role of *Delta*, we genetically perturbed the early expression of *Delta* in the larval imaginal disc. We used the publicly available Flylight Image Database (50) to select GAL4 drivers that are active in the lateral male genital primordium of the larval disc and carried out a screen with the goal of knocking down *Delta* sufficiently early to specifically disrupt the early ancestral function. From this UAS-GAL4 *Delta*-RNAi screen, we successfully identified a GAL4 driver in the *MSR1* locus (GMR64C05-GAL4) that knocked down *Delta* in the lateral clusters of the male genital primordium without affecting *Delta*'s expression in other regions (Fig. 4, C to E). As a result, 100% of the early *Delta* knockdown animals exhibited defects in genital eversion (Fig. 4G and fig. S11C). At a pupal time point when the precursor of the lateral plates and claspers has everted in the control animals (Fig. 4F), the genitalia of *Delta* knockdown animals were still encapsulated within the tissue that

should form the eighth body segment (Fig. 4G). Thirty-two percent of male adults completely lacked external genitalia, which we postulate is due to failure of eversion (fig. S11, A and B). Because RNAi is often incomplete and variable, *Delta* was not completely knocked down in some animals. Nonetheless, the adults that formed external genitalia had defects in either claspers, lateral plates, or posterior lobes, or all three structures (Fig. 4, H to J, and fig. S11D). These results indicate that *Delta* had an early function important for genital disc eversion, a complex process that is presumably necessary for the genital development of all *Drosophila* species and predates the evolution of the posterior lobe.

Delta is upstream of a vast apical extracellular matrix network

Our previous work on the cellular development of the posterior lobe revealed that the cells of the posterior lobe drastically increase in height to project from the developing lateral plate and implicated an important role for the apical extracellular matrix (aECM) in shaping the posterior lobe cells (30). An ancestral aECM network that includes the zona pellucida domain-containing protein Dumpy is associated with the developing genitalia of both lobed and nonlobed species. It forms physical associations with conserved structures in the center of the genitalia, namely, the hypandrium, phallus, and the sheath, as well as the surrounding structures, such as the clasper. In nonlobed species, the aECM forms weak connections to the space between the lateral plate and clasper, while it has expanded to form prominent connections to the posterior lobe in *D. melanogaster*, where it is essential for proper posterior lobe development (30). It is speculated that Dumpy provides structural support as the intrinsic factors of the posterior lobe cells drive elongation, or that Dumpy tethers create a mechanical force to pull the cells of the posterior lobe (30). Given that Dumpy is a cellular effector molecule and likely downstream of the posterior lobe GRN, we were curious to know whether there is a potential link between *Delta* and *dumpy*. Similar to *Delta*, *dumpy* expression is spatially expanded in *D. melanogaster* compared to nonlobed species, resembling the Notch-responsive regions (30). Therefore, we examined whether Dumpy is downstream of *Delta* and whether it also has an early role in genital eversion. First, using a line in which the Dumpy protein is endogenously tagged with a yellow fluorescent protein (Dumpy:YFP) (52, 53), we detected Dumpy deposition within the epithelial folds of the larval genital disc (Fig. 4K). Dumpy deposits within the genital disc clearly line the surface of the male genital primordium in the lateral regions, overlaying the *Delta*-expressing epithelial cells (Fig. 4K and fig. S12B). To test whether Dumpy has a role in the *Delta*-dependent genital disc eversion process, we examined whether Dumpy is disrupted under larval disc *Delta* RNAi conditions. In *Delta* RNAi animals with an eversion defect, Dumpy lost clear associations with the apical surface of the everting cells compared to the control that form concentrated deposits (Fig. 4, L and M). Dumpy deposition in the larval genital disc was also decreased compared to controls (fig. S12B), suggesting that the early activity of *Delta* is necessary for the presence of Dumpy at the surface of the male genital primordium. Together, these data suggest that Dumpy acts downstream of *Delta* during the eversion of the genital disc. In *Delta* RNAi conditions where the animals escaped the defective genital disc eversion phenotype, Dumpy localization was decreased on the prospective lobe-forming regions of the lateral plates (fig. S14), suggesting that the novel role of Dumpy in posterior lobe development is also downstream of *Delta*.

DISCUSSION

Despite strong implications for the role of signaling pathways in the formation of many novel morphologies, efforts to understand how their pivotal roles were established have lagged far behind. Here, we identified a signal source, *Delta*, that is crucial to a novelty that formed on an intermediate timescale, permitting an opportune glance at its developmental and evolutionary past. Our comparative analysis of lobed and nonlobed species revealed that the *Delta* signal predated the posterior lobe novelty, suggesting roles preceding its evolution. The *Delta* signal initiates at a remarkably early developmental time, where it is required for the fundamental process of disc eversion, common to all species analyzed in this study. We placed this role in eversion upstream of the terminal effector *Dumpy*, which also participates directly in posterior lobe development. These findings suggest that morphological novelties that originate on macroevolutionary scales may evolve through elaborations of ancestral signaling centers whose beginnings may have been obscured by the passage of time. Because of their antiquity, the genetic architecture of these ancestral systems may be quite intricate and robust. We discuss below the implications of such “ancestral complexity” in understanding the origins of elaborate morphological structures.

The phenotypes we observed for *Delta* knockdown are some of the most marked posterior lobe defects we have found, and overexpression of the Notch intracellular domain is the only treatment currently known that markedly increases lobe size to date [but also see (54)]. Considering the expanded expression of *Delta* near the developing lobe of *D. melanogaster*, this motivated our search for enhancers to trace *Delta*’s evolutionary history. While a combination of cis and trans differences accounts for the expanded expression we observed, the reporters notably allowed us to lineage trace the developmental trajectory of this signaling center back to the third instar genital disc. Before this study, our knowledge of the morphogenetic processes that occur during early genital disc development at pupal stages was sparse. Valuable studies in larval genital discs have identified genes and signaling pathways involved in anterior and posterior compartmentalization and patterning (55–57), and a recent study by Ridgway *et al.* (54) showed differences in *Sox21b* expression in the genital disc that underlie changes in posterior lobe size between *Drosophila simulans* and *Drosophila mauritiana*. While the eversion of the genital disc had previously been described (51), the molecular mechanisms governing this phenomenon remained largely uninvestigated. Our results highlight how novelties may depend upon signals initially deployed to pattern ancestral features at earlier stages of development. Work in the bat limb also hints at this possibility with the reinitiation of *Shh* signaling (13) specific to the bat wing after expression of *Shh* in the zone of polarizing activity has ceased. Such findings underline the importance of considering a wide range of developmental times that may bear upon evolving traits, rather than focusing solely on stages proximate to the developmental manifestation of the trait.

Our survey of *Delta* regulatory elements also revealed a redundant architecture that controls a broad domain of ligand expression that shifts over time. Redundant enhancers are quite common and are thought to foster robustness to environmental and genetic variation (58–60). The enhancer elements we found in the genital disc cover a wider spatial domain than was observed for the pupal enhancers (fig. S10). This suggests that the lobe-patterning signal center may have emerged from a portion of the total genital disc signaling center. Furthermore, our *Delta* knockdown experiments

revealed a counterintuitive relationship between *Delta* expression and lobe patterning. When *Delta* RNAi is driven by the early *Delta* enhancer, the ventral expansion is completely ablated, but lobe morphology is unaffected (fig. S9, D to F). This suggests that the dorsal expansion (near the anal plate) represents the portion of the *Delta* signal relevant to the formation of this novelty. These results underscore the importance of fine-scale genetic manipulations to localize which portion of an expression pattern is most necessary to the developmental events under study.

We previously implicated a critical role of the aECM in the development of the posterior lobe (30). Here, we demonstrated that *Dumpy* deposition was affected when *Delta* was knocked down early in the larval genital disc, suggesting that *Dumpy* serves an early *Delta*-dependent role in disc eversion and a late role downstream of *Delta* in the posterior lobe network. A plausible model is that *dumpy* is regulated by *Delta* in the larval genital disc of both lobed and nonlobed species, carrying out an ancestral function in genital disc eversion. As *Delta* evolved an expanded expression pattern in *D. melanogaster*, the expression of *dumpy* may have expanded correspondingly. According to this model, a preexisting target of *Delta* became involved in the development of the posterior lobe by becoming active in a broader expression domain (Fig. 5). Crucial to examining this model is identifying the regulatory elements that govern *dumpy* expression in the larval genital disc as well as the posterior lobe. It is also possible, if not likely, that in addition to deploying preexisting downstream targets, the Notch pathway has also

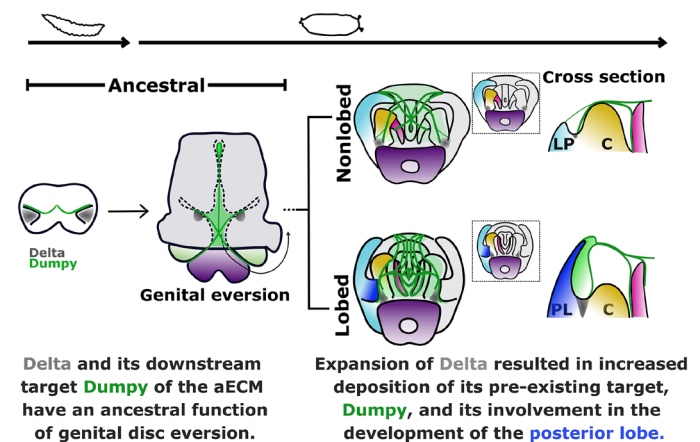


Fig. 5. A Notch signal required for a morphological novelty in *Drosophila* has antecedent functions in genital disc eversion. (From left to right) A *Delta*/Notch signaling center (gray) is active in the larval genital disc with an ancestral function in genital disc eversion, a process that is conserved among *Drosophila* species. *Dumpy* (green lines), a component of the aECM, is associated with antecedent roles in genital disc eversion in a *Delta*-dependent manner and may be a preexisting target of *Delta*. The *Delta*/Notch signaling center persists through pupal development in nonlobed species (top right), where it carries out potential ancestral roles in later pupal development, such as lateral plate (LP, light blue) and clasper (C, yellow) development through *Dumpy* connections to the epithelia of these structures. In lobe-forming species (bottom right), the ancestral pattern at the base of the lateral plate/clasper precursor (shaded green) later spatially expands along the two structures through trans regulatory changes (lateral plate in light blue, clasper in yellow) as the posterior lobe (royal blue) forms. This model suggests that as the expanded expression of *Delta* evolved, its preexisting downstream target, *Dumpy*, gained an additional activity in posterior development. Insets show cartoon genitalia without *dumpy* depictions.

gained novel targets in lobed species that contribute to posterior lobe development.

Our previous work on the evolution of the posterior lobe had shown that an ancestral network of the embryonic posterior spiracle was co-opted into genital development, where it is required for posterior lobe development (22). The connection between the Delta signaling center and co-opted posterior spiracle network, however, is yet to be explored. An interesting question is whether the expansion of Delta is downstream of the co-opted network or vice versa. This turns out to be a complicated question to resolve. Our preliminary data suggest that the expansion of Delta during pupal stages is downstream of *Poxn*, a key component of the co-opted network (fig. S15). In a *Poxn* null mutant, Delta expression is reduced to a small zone of expression that resembles the ancestral expression of Delta in nonlobed species (fig. S15). This *Poxn*-independent activity of Delta is in agreement with our data showing that Delta has a much earlier onset of expression than *Poxn*. *Poxn* itself (and many components of the spiracle network) is activated in two waves during development of tissues near the posterior lobe: namely, at the lateral-plate-clasper junction and then in the posterior lobe itself. Although *Poxn* mutants have lower Delta expression and do not form proper posterior lobes, we suspect that the *Poxn*-dependent expression of Delta is due to the expression of *Poxn* during the first wave of its expression, which corresponds spatially and temporally with Delta expression that occurs adjacent to the developing posterior lobe.

We postulate that the posterior spiracle network co-option event mostly involved intermediate transcription factors within the posterior lobe network. Knockdown of most of the terminal effector genes of the posterior spiracle network had very mild phenotypes compared to the transcriptional regulators we found in the network, suggesting that deployment of key genes such as *Poxn* during posterior lobe development were important to the posterior lobe's origin. We may, however, find terminal effectors of the posterior spiracle network that have greater phenotypic impacts on the posterior lobe. Of note, *dumpy* is expressed in the posterior spiracle as well (61) and determining whether or not these activities were co-opted will require an in-depth analysis of the *dumpy* regulatory region. Last, the fact that Delta is expressed so early suggests that this pattern and its elaboration predated the co-option of the spiracle network because the co-opted genes are active at much later developmental stages, as the posterior lobe is emerging. Thus, it might be possible that the co-option of the spiracle network contributed to the spatial expansion of Delta expression, though it is entirely possible that these two events are unrelated and occurred independently.

The complexity of ancestral systems has been an important concept in developmental evolution. As we learned from sequencing the human genome (62) and an increasing array of basally branching organisms (63–65), the field realized that most transcription factors and signaling pathways are conserved within animals and beyond, rather than each clade having an abundance of lineage specific regulatory genes and gene families. Our work here causes us to appreciate how ancestrally complex signaling centers may apply to newly formed novelties. These centers may show faint signs of relation to signals deployed much earlier in development. The regulatory architecture of their participating loci may show unexpected intricacies, such as redundancy that resulted from the evolutionary refinement of their ancestral roles. Our efforts to resolve these faint connections and complex developmental systems are crucial to

developing a sophisticated understanding of what would otherwise appear to have arisen through inexplicable events.

MATERIALS AND METHODS

Drosophila strains and husbandry

Fly stocks were reared at room temperature on standard cornmeal agar media. For RNAi experiments, flies were reared at 29°C. The *D. melanogaster* line used in this study is mutant for the *yellow* and *white* genes and was isogenized for eight generations (y^1w^1 , Bloomington *Drosophila* Stock Center #1495). Outgroup species that lack a posterior lobe (*D. ananassae* #0000-1005.01 and *D. biarmipes* #0000-1028.01) were obtained from the University of California, San Diego *Drosophila* Stock Center (now called The National *Drosophila* Species Stock Center at Cornell University). All stocks are listed in table S1.

Transgenic constructs

To generate GFP reporter flies, regulatory elements were polymerase chain reaction (PCR) amplified using primers listed in table S2, and cloned into a vector containing GFP and a minimal hsp70 promoter (pS3aG) (66). Primers were designed using the GenePalette software tool (67). Asc I and Sbf I restriction sites were added to the primer sequences (Integrated DNA Technologies) to insert the amplified region into the multiple cloning site of the vector. Regions of interest were amplified from genomic DNA prepared by the DNeasy Blood & Tissue Kit (QIAGEN). *D. melanogaster* transformant lines were created by phiC31-mediated site-specific recombination into the 68A4 “attP2” landing site on the third chromosome (68) or the 51D site on the second chromosome (69) by Rainbow Transgenics. For each GFP reporter, two to five independent insertion lines were analyzed.

Larval and pupal genital sample preparation

Pupal genital samples were prepared for in situ hybridization according to Glassford *et al.* (22, 70). In short, to standardize aging, male white prepupae were incubated at 25°C until ready for dissection. Pupae were cut in half in cold phosphate-buffered saline (PBS), fat bodies were flushed out, and the pupal case was removed. Samples younger than 20 hours APF and larval samples were cut in half and turned inside out at the posterior end to remove fat bodies to prevent damaging or dislodging the delicate early genitalia. All samples were fixed in PBS with 0.1% Triton X-100 and 4% paraformaldehyde (PBT-fix) for 30 min at room temperature. Samples containing fluorescent labels or being prepared for immunostaining were then washed twice in PBT. Samples to be used for in situ hybridization were rinsed twice in methanol and twice in ethanol, and stored at –20°C in 100% ethanol.

Generation of CRISPR-Cas9 mutants

Deletion of the enhancers of *Delta* was accomplished by CRISPR-Cas9 homology-directed repair using guide RNA (gRNA) targets flanking the enhancer boundaries (table S2). For the deletion of enhancer1, *D. melanogaster* embryos were injected by Rainbow Transgenics with a mixture of nos-Cas9 vector (250 ng/μl), 100 ng/μl of each gRNA vector (pCFD3-dU63gRNA), and donor plasmid (500 ng/μl). The donor plasmids for homology-directed repair contain a 3XP3::DsRed cassette flanked by approximately 1 kb of genomic DNA. Transformants were identified by the expression of DsRed in the eyes of the progeny

of the injected flies. Embryo injections to delete the early enhancer of *Delta* were performed in-house. A mixture of 200 ng/μl per single guide RNA (sgRNA) and donor vector (500 ng/μl) was injected into nos-Cas9-expressing embryos (Bloomington #78781). sgRNAs were generated by in vitro transcription. Briefly, a double-stranded DNA template containing a T7 promoter was amplified by PCR. In vitro transcription was then carried out using the MEGAscript T7 Transcription kit (Invitrogen).

In situ hybridization

To detect mRNA localization, in situ hybridization was performed following the protocol described in Glassford *et al.* (22), using the InsituPro VSi robot (Intavis Bioanalytical Instruments) (70). Briefly, fixed samples were washed in methanol, rehydrated in PBT (PBS with 0.1% Triton X-100), fixed in PBT-fix (PBT containing 4% para-formaldehyde), and incubated in hybridization buffer for 1 hour at 65°C. Prehybridized samples were then incubated with digoxigenin riboprobes (primers for amplifying mRNA probes listed in table S2) for 16 hours at 65°C and subsequently washed in hybridization buffer followed by PBT washes to remove unbound riboprobes. To reduce background noise, samples were blocked in PBT with 1% bovine serum albumin for 2 hours. Blocked samples were then incubated with anti-digoxigenin antibody Fab fragments conjugated to alkaline phosphatase (Roche Diagnostics) diluted in PBT at 1:6000 overnight at 4°C. After several PBT washes, alkaline phosphatase color reactions were performed by incubating samples in nitro-blue tetrazolium chloride and 5-bromo-4-chloro-3'-indolylphosphate p-toluidine salt (NBT/BCIP) (Promega) and monitored under a dissecting microscope. Color reactions were stopped by PBT washes when purple stain was detected. Samples were mounted on a glass slide coated with poly-L-lysine in an 80% glycerol 0.1 M tris-HCl (pH 8.0) solution. Hybridization chain reaction experiments were carried out with Molecular Instruments HCR3.0 reagents using the protocol for fruit fly embryos.

Immunostaining

To detect the expression patterns of proteins, genital samples removed from the pupal membrane were incubated in primary antibody diluted in PBT at 4°C overnight. The following primary antibodies were used: monoclonal mouse anti-Delta 1:100 (DSHB, #C594.9B0s), polyclonal goat anti-Delta 1:100 (dL-19, Santa Cruz Biotechnology Inc.), and rat anti-Ecadherin 1:100 (DSHB, #DCAD2). To remove unbound primary antibody, the samples were washed in PBT three to five times over the course of an hour. To detect bound primary antibody, the samples were subsequently incubated in a fluorescent dye-conjugated secondary antibody diluted in PBT and incubated either at 4°C overnight or at room temperature for 4 hours. The following secondary antibodies were used: donkey anti-mouse Alexa 488 1:400 (A21202, Thermo Fisher Scientific), donkey anti-goat Cy2 1:400 (705-225-147, Jackson ImmunoResearch), and donkey anti-rat Alexa 594 1:400 (A21209, Invitrogen). Samples were then washed in PBT, incubated in 50% PBT and 50% glycerol solution, and mounted on a glass slide in an 80% glycerol 0.1 M tris-HCl (pH 8.0) solution. Mounting wells were made with a single layer of double sticky tape on the glass slides. To avoid rotation of the sample during mounting, the wells were coated with poly-L-lysine solution. Glass coverslips were placed on the samples to seal the wells.

Microscopy and image analysis

Cuticles of adult genitalia and stained in situ hybridization samples were imaged on a Leica DM2000 with a Leica DFC540 camera at 20× magnification. Fluorescently labeled samples were imaged using an Olympus Fluoview 1000 or a Leica TCS SP8 confocal microscope at 40× magnification. Larval samples were imaged at 63× magnification where indicated. Images were processed with Fiji (71).

Supplementary Materials

This PDF file includes:

Figs. S1 to S15

Tables S1 and S2

REFERENCES AND NOTES

1. D. Rudel, R. J. Sommer, The evolution of developmental mechanisms. *Dev. Biol.* **264**, 15–37 (2003).
2. M. Rubinstein, F. S. J. De Souza, Evolution of transcriptional enhancers and animal diversity. *Philos. Trans. R. Soc. B* **368**, 20130017 (2013).
3. S. B. Carroll, J. K. Grenier, S. D. Weatherbee, *From DNA to Diversity: Molecular Genetics and the Evolution of Animal Design*, Blackwell Science, (2005).
4. J. Gerhart, M. Kirschner, The theory of facilitated variation. *Proc. Natl. Acad. Sci. U.S.A.* **104**, 8582–8589 (2007).
5. S. B. Carroll, Evo-Devo and an expanding evolutionary synthesis: A genetic theory of morphological evolution. *Cell* **134**, 25–36 (2008).
6. A. Pires-dasilva, R. J. Sommer, The evolution of signalling pathways in animal development. *Nat. Rev. Genet.* **4**, 39–49 (2003).
7. N. Perrimon, C. Pitsoulis, B.-Z. Shilo, Signaling mechanisms controlling cell fate and embryonic patterning. *Cold Spring Harb. Perspect. Biol.* **4**, a005975 (2012).
8. S. Kuraku, R. Usuda, S. Kuratani, Comprehensive survey of carapacial ridge-specific genes in turtle implies co-option of some regulatory genes in carapace evolution. *Evol. Dev.* **7**, 3–17 (2005).
9. G. A. Loreda, A. Brukman, M. P. Harris, D. Kagle, E. E. Leclair, R. Gutman, E. Denney, E. Henkelman, B. P. Murray, J. F. Fallon, R. S. Tuan, S. F. Gilbert, Development of an evolutionarily novel structure: Fibroblast growth factor expression in the carapacial ridge of turtle embryos. *J. Exp. Zool.* **291**, 274–281 (2001).
10. S. B. Carroll, J. Gates, D. N. Keys, S. W. Paddock, G. E. F. Panganiban, J. E. Selegue, J. A. Williams, Pattern formation and eyespot determination in butterfly wings. *Science* **265**, 109–114 (1994).
11. D. N. Keys, D. L. Lewis, J. E. Selegue, B. J. Pearson, L. V. Goodrich, R. L. Johnson, J. Gates, M. P. Scott, S. B. Carroll, Recruitment of a hedgehog regulatory circuit in butterfly eyespot evolution. *Science* **283**, 532–534 (1999).
12. K. E. Sears, Molecular determinants of bat wing development. *Cells Tissues Organs* **187**, 6–12 (2008).
13. D. Hockman, C. J. Cretekos, M. K. Mason, R. R. Behringer, D. S. Jacobs, N. Illing, A second wave of Sonic hedgehog expression during the development of the bat limb. *Proc. Natl. Acad. Sci. U.S.A.* **105**, 16982–16987 (2008).
14. S. D. Weatherbee, R. R. Behringer, J. J. Rasweiler IV, L. A. Niswander, Interdigital webbing retention in bat wings illustrates genetic changes underlying amniote limb diversification. *Proc. Natl. Acad. Sci. U.S.A.* **103**, 15103–15107 (2006).
15. M. P. Harris, J. F. Fallon, R. O. Prum, Shh-Bmp2 signaling module and the evolutionary origin and diversification of feathers. *J. Exp. Zool.* **294**, 160–176 (2002).
16. C. M. Clark-hachtel, Y. Tomoyasu, Two sets of candidate crustacean wing homologues and their implication for the origin of insect wings. *Nat. Ecol. Evol.* **4**, 1694–1702 (2020).
17. A. P. Moczek, L. M. Nagy, Diverse developmental mechanisms contribute to different levels of diversity in horned beetles. *Evol. Dev.* **185**, 175–185 (2005).
18. M. P. Harris, S. Williamson, J. F. Fallon, H. Meinhardt, R. O. Prum, Molecular evidence for an activator-inhibitor mechanism in development of embryonic feather branching. *Proc. Natl. Acad. Sci. U.S.A.* **102**, 11734–11739 (2005).
19. O. A. Tarazona, D. H. Lopez, L. A. Slota, M. J. Cohn, Evolution of limb development in cephalopod mollusks. *eLife* **8**, e43828 (2019).
20. D. J. Emlen, Q. Szafran, L. S. Corley, I. Dworkin, Insulin signaling and limb-patterning: Candidate pathways for the origin and evolutionary diversification of beetle 'horns'. *Heredity* **97**, 179–191 (2006).
21. B. M. I. Vreede, J. A. Lynch, S. Roth, É. Sucena, Co-option of a coordinate system defined by the EGFR and Dpp pathways in the evolution of a morphological novelty. *Evodevo* **4**, 7–12 (2013).
22. W. J. Glassford, W. C. Johnson, N. R. Dall, S. J. Smith, Y. Liu, W. Boll, M. Noll, M. Rebeiz, Co-option of an ancestral Hox-regulated network underlies a recently evolved

- morphological novelty article co-option of an ancestral Hox-regulated network underlies a recently evolved morphological novelty. *Dev. Cell* **34**, 520–531 (2015).
23. A. Vargas-Lowman, D. Armisen, C. F. Burguez Floriano, I. da Rocha Silva Cordeiro, S. Viala, M. Bouchet, M. Bernard, A. Le Bouquin, M. E. Santos, A. Berlioz-Barbier, A. Salvador, F. F. Figueiredo Moreira, F. Bonneton, A. Khila, Cooption of the pteridine biosynthesis pathway underlies the diversification of embryonic colors in water striders. *Proc. Natl. Acad. Sci. U.S.A.* **116**, 19046–19054 (2019).
 24. A. Martin, R. Papa, N. J. Nadeau, R. I. Hill, B. A. Counterman, G. Halder, C. D. Jiggins, M. R. Kronforst, A. D. Long, W. O. McMillan, R. D. Reed, Diversification of complex butterfly wing patterns by repeated regulatory evolution of a Wnt ligand. *Proc. Natl. Acad. Sci. U.S.A.* **109**, 12632–12637 (2012).
 25. A. Mazo-Vargas, C. Concha, L. Livraghi, D. Massardo, R. W. R. Wallbank, L. Zhang, J. D. Papador, D. Martinez-Najera, C. D. Jiggins, M. R. Kronforst, C. J. Breuker, R. D. Reed, N. H. Patel, W. O. McMillan, A. Martin, Macroevolutionary shifts of WntA function potentiate butterfly wing-pattern diversity. *Proc. Natl. Acad. Sci. U.S.A.* **114**, 10701–10706 (2017).
 26. L. Livraghi, J. J. Hanly, S. M. Van Bellghem, G. Montejó-Kovacevich, E. S. M. van der Heijden, L. S. Loh, A. Ren, I. A. Warren, J. J. Lewis, C. Concha, L. Hebberecht, C. J. Wright, J. M. Walker, J. Foley, Z. F. Goldberg, H. Arenas-Castro, C. Salazar, M. W. Perry, R. Papa, A. Martin, W. O. McMillan, C. D. Jiggins, Cortex cis-regulatory switches establish scale colour identity and pattern diversity in heliconius. *eLife* **10**, e68549 (2021).
 27. T. Werner, S. Koshikawa, T. M. Williams, S. B. Carroll, Generation of a novel wing colour pattern by the Wingless morphogen. *Nature* **464**, 1143–1148 (2010).
 28. L. Arnoult, K. F. Y. Su, D. Manoel, C. Minervino, J. Magriña, N. Gompel, B. Prud'homme, Emergence and diversification of fly pigmentation through evolution of a gene regulatory module. *Science* **339**, 1423–1426 (2013).
 29. C. R. Brunetti, J. E. Selegue, A. Monteiro, V. French, P. M. Brakefield, S. B. Carroll, The generation and diversification of butterfly eyespot color patterns. *Curr. Biol.* **11**, 1578–1585 (2001).
 30. S. J. Smith, L. A. Davidson, M. Rebeiz, Evolutionary expansion of apical extracellular matrix is required for the elongation of cells in a novel structure. *eLife* **9**, e55965 (2020).
 31. M. Levine, E. H. Davidson, Gene regulatory networks for development. *Proc. Natl. Acad. Sci. U.S.A.* **102**, 4936–4942 (2005).
 32. S. N. Murugesan, H. Connahs, Y. Matsuo, M. Das Gupta, G. J. L. Tiong, M. Huq, V. Gowri, S. Monroe, K. D. Deem, T. Werner, Y. Tomoyasu, A. Monteiro, Butterfly eyespots evolved via cooption of an ancestral gene-regulatory network that also patterns antennae, legs, and wings. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2108661119 (2022).
 33. E. Preger-Ben Noon, G. Sabaris, D. M. Ortiz, J. Sager, A. Liebowitz, D. L. Stern, N. Frankel, Comprehensive analysis of a cis-regulatory region reveals pleiotropy in enhancer function. *Cell Rep.* **22**, 3021–3031 (2018).
 34. W. G. Eberhard, Evolution of genitalia: Theories, evidence, and new directions. *Genetica* **138**, 5–18 (2010).
 35. A. Kopp, J. R. True, Evolution of male sexual characters in the oriental *Drosophila melanogaster* species group. *Evol. Dev.* **4**, 278–291 (2002).
 36. S. Jagadeeshan, R. S. Singh, A time-sequence functional analysis of mating behaviour and genital coupling in *Drosophila*: Role of cryptic female choice and male sex-drive in the evolution of male genitalia. *J. Evol. Biol.* **19**, 1058–1070 (2006).
 37. M. Polak, A. J. Moehring, No evidence for external genital morphology affecting cryptic female choice and reproductive isolation in *Drosophila*. *Evolution* **69**, 1797–1807 (2015).
 38. S. R. Frazee, J. P. Masly, Multiple sexual selection pressures drive the rapid evolution of complex morphology in a male secondary genital structure. *Ecol. Evol.* **5**, 4437–4450 (2015).
 39. G. Rice, J. R. David, Y. Kamimura, J. P. Masly, A. P. McGregor, O. Nagy, S. Noselli, M. D. S. Nunes, P. O'Grady, E. Sánchez-Herrero, M. L. Siegal, M. J. Toda, M. Rebeiz, V. Courtier-Argogozo, A. Yassin, A standardized nomenclature and atlas of the male terminalia of *Drosophila melanogaster*. *Fly* **13**, 51–64 (2019).
 40. S. Kumar, G. Stecher, M. Suleski, S. Blair Hedges, TimeTree: A resource for timelines, timetrees, and divergence times. *Mol. Biol. Evol.* **34**, 1812–1819 (2017).
 41. W. Boll, M. Noll, The *Drosophila* Pox neuro gene: Control of male courtship behavior and fertility as revealed by a complete dissection of all enhancers. *Development* **129**, 5667–5681 (2002).
 42. E. Zacharioudaki, S. J. Bray, Tools and methods for studying Notch signaling in *Drosophila melanogaster*. *Methods* **68**, 173–182 (2014).
 43. M. J. Go, D. S. Eastman, S. Artavanis-tsakonas, Cell proliferation control by Notch signaling in *Drosophila* development. *Development* **125**, 2031–2040 (1998).
 44. D. Henrique, F. Schweisguth, Mechanisms of Notch signaling: A simple logic deployed in time and space. *Development* **146**, dev172148 (2019).
 45. S. Yamamoto, K. L. Schulze, H. J. Bellen, Introduction to Notch signaling. *Methods Mol Biol* **1187**, 1–14 (2014).
 46. D. T. Nellesen, E. C. Lai, J. W. Posakony, Discrete enhancer elements mediate selective responsiveness of Enhancer of split complex genes to common transcriptional activators. *Dev. Biol.* **213**, 33–53 (1999).
 47. B. D. Pfeiffer, A. Jenett, A. S. Hammonds, T. B. Ngo, S. Misra, C. Murphy, A. Scully, J. W. Carlson, K. H. Wan, T. R. Lavery, C. Mungall, R. Svirskas, J. T. Kadonaga, C. Q. Doe, M. B. Eisen, S. E. Celniker, G. M. Rubin, Tools for neuroanatomy and neurogenetics in *Drosophila*. *Proc. Natl. Acad. Sci. U.S.A.* **105**, 9715–9720 (2008).
 48. L. He, R. Binari, J. Huang, J. Faló-Sanjuan, N. Perrimon, In vivo study of gene expression with an enhanced dual-color fluorescent transcriptional timer. *eLife* **8**, e46181 (2019).
 49. J. M. Tennesen, C. S. Thummel, Coordinating growth and maturation—Insights from *Drosophila*. *Curr. Biol.* **21**, PR750–PR757 (2011).
 50. J. Aurélie, C. Estella, M. W. Giorgianni, M. Slattery, T. R. Lavery, G. M. Rubin, R. S. Mann, Resource a survey of 6,300 genomic fragments for cis-regulatory activity in the imaginal discs of *Drosophila melanogaster*. *Cell Rep.* **2**, 1014–1024 (2012).
 51. F. Epper, The evagination of the genital imaginal discs of *Drosophila melanogaster*. *Development* **192**, 280–284 (1983).
 52. C. M. Lye, H. W. Naylor, B. Sanson, Subcellular localisations of the CPTI collection of YFP-tagged proteins in *Drosophila* embryos. *Development* **141**, 4006–4017 (2014).
 53. N. Lowe, J. S. Rees, J. Roote, E. Ryder, I. M. Armean, G. Johnson, E. Drummond, H. Spriggs, J. Drummond, J. P. Magbanua, H. Naylor, R. Bastock, S. Huelsmann, V. Trovisco, M. Landgraf, S. Knowles-Barley, J. D. Armstrong, H. White-cooper, C. Hansen, R. G. Phillips, UK *Drosophila* Protein Trap Screening Consortium, S. Consortium, K. S. Lilley, S. Russell, D. S. Johnston, Analysis of the expression patterns, subcellular localisations and interaction partners of *Drosophila* proteins using a pigP protein trap library. *Development* **141**, 3994–4005 (2014).
 54. A. M. Ridgway, E. J. Hood, J. F. Jimenez, M. D. S. Nunes, A. P. McGregor, A. M. Ridgway, E. J. Hood, J. F. Jimenez, M. D. S. Nunes, A. P. McGregor, Sox21b underlies the rapid diversification of a novel male genital structure between *Drosophila* species. *Curr. Biol.* **34**, 1114–1121.e7 (2024).
 55. E. H. Chen, B. S. Baker, Compartmental organization of the *Drosophila* genital imaginal discs. *Development* **124**, 205–218 (1997).
 56. L. Sánchez, F. Casares, N. Gorfinkel, I. Guerrero, The genital disc of *Drosophila melanogaster*: II. Role of the genes hedgehog, decapentaplegic and wingless. *Dev. Genes Evol.* **207**, 229–241 (1997).
 57. B. Estrada, F. Casares, E. Sánchez-Herrero, Development of the genitalia in *Drosophila melanogaster*. *Differentiation* **71**, 299–310 (2003).
 58. S. Barolo, Shadow enhancers: Frequently asked questions about distributed cis-regulatory information and enhancer redundancy. *Bioessays* **34**, 135–141 (2012).
 59. M. W. Perry, A. N. Boettiger, J. P. Bothma, M. Levine, Shadow enhancers foster robustness of *Drosophila* gastrulation. *Curr. Biol.* **20**, 1562–1567 (2010).
 60. N. Frankel, G. K. Davis, D. Vargas, S. Wang, F. Payre, D. L. Stern, Phenotypic robustness conferred by apparently redundant transcriptional enhancers. *Nature* **466**, 490–493 (2010).
 61. M. B. Wilkin, M. N. Becker, D. Mulvey, I. Phan, A. Chao, K. Cooper, H. J. Chung, I. D. Campbell, M. Baron, R. MacIntyre, *Drosophila* Dumpy is a gigantic extracellular protein required to maintain tension at epidermal-cuticle attachment sites. *Curr. Biol.* **10**, 559–567 (2000).
 62. J. C. Venter, M. D. Adams, E. W. Myers, P. W. Li, R. J. Mural, G. G. Sutton, H. O. Smith, M. Yandell, C. A. Evans, R. A. Holt, J. D. Gocayne, P. Amanatides, R. M. Ballew, D. H. Huson, J. R. Wortman, Q. Zhang, C. D. Kodira, X. H. Zheng, L. Chen, M. Skupski, G. Subramanian, P. D. Thomas, J. Zhang, G. L. G. Miklos, C. Nelson, S. Broder, A. G. Clark, J. Nadeau, V. A. McKusick, N. Zinder, A. J. Levine, R. J. Roberts, M. Simon, C. Slayman, M. Hunkapiller, R. Bolanos, A. Delcher, I. Dew, D. Fasulo, M. Flanigan, L. Florea, A. Halpern, S. Hannenhalli, S. Kravitz, S. Levy, C. Mobarry, K. Reinert, K. Remington, J. Abu-Threideh, E. Beasley, K. Biddick, V. Bonazzi, B. Brandon, M. Cargill, I. Chandramouliswaran, R. Charlab, K. Chaturvedi, Z. Deng, V. di Francesco, P. Dunn, K. Eilbeck, C. Evangelista, A. E. Gabrielian, W. Gan, W. Ge, F. Gong, Z. Gu, P. Guan, T. J. Heiman, M. E. Higgins, R. R. Ji, Z. Ke, K. A. Ketchum, Z. Lai, Y. Lei, Z. Li, J. Li, Y. Liang, X. Lin, F. Lu, G. V. Merkulov, N. Milshina, H. M. Moore, A. K. Naik, V. A. Narayan, B. Neelam, D. Nusskern, D. B. Rusch, S. Salzberg, W. Shao, B. Shue, J. Sun, Z. Y. Wang, A. Wang, X. Wang, J. Wang, M. H. Wei, R. Wides, C. Xiao, C. Yan, A. Yao, J. Ye, M. Zhan, W. Zhang, H. Zhang, Q. Zhao, L. Zheng, F. Zhong, W. Zhong, S. C. Zhu, S. Zhao, D. Gilbert, S. Baumhueter, G. Spier, C. Carter, A. Cravchik, T. Woodage, F. Ali, H. An, A. Awe, D. Baldwin, H. Baden, M. Barnstead, I. Barrow, K. Beeson, D. Busam, A. Carver, A. Center, M. L. Cheng, L. Curry, S. Danaher, L. Davenport, R. Desilets, S. Dietz, K. Dodson, L. Doup, S. Ferriera, N. Garg, A. Gluecksmann, B. Hart, J. Haynes, C. Haynes, C. Heiner, S. Hladun, D. Hostin, J. Houck, T. Howland, C. Ibbegwam, J. Johnson, F. Kalush, L. Kline, S. Koduru, A. Love, F. Mann, D. May, S. McCawley, T. McIntosh, I. McMullen, M. Moy, L. Moy, B. Murphy, K. Nelson, C. Pfannkuch, E. Pratt, V. Puri, H. Qureshi, M. Reardon, R. Rodriguez, Y. H. Rogers, D. Romblad, B. Ruhfel, R. Scott, C. Sitter, M. Smallwood, E. Stewart, R. Strong, E. Suh, R. Thomas, N. N. Tint, S. Tse, C. Vech, G. Wang, J. Wetter, S. Williams, M. Williams, S. Windsor, E. Winn-Deen, K. Wolfe, J. Zaveri, K. Zaveri, J. F. Abril, R. Guigo, M. J. Campbell, K. V. Sjolander, B. Karlak, A. Kejariwal, H. Mi, B. Lazareva, T. Hatton, A. Narechania, K. Diemer, A. Muruganujan, N. Guo, S. Sato, Y. Bafna, S. Istrail, R. Lipert, R. Schwartz, B. Walenz, S. Yooseph, D. Allen, A. Basu, J. Baxendale, L. Blick, M. Caminha, J. Carnes-Stine, P. Caulk, Y. H. Chiang, M. Coyne, C. Dahlke, A. D. Mays,

- M. Dombroski, M. Donnelly, D. Ely, S. Esparham, C. Fosler, H. Gire, S. Glanowski, K. Glasser, A. Glodek, M. Gorokhov, K. Graham, B. Gropman, M. Harris, J. Heil, S. Henderson, J. Hoover, D. Jennings, C. Jordan, J. Jordan, J. Kasha, L. Kagan, C. Kraft, A. Levitsky, M. Lewis, X. Liu, J. Lopez, D. Ma, W. Majoros, J. McDaniel, S. Murphy, M. Newman, T. Nguyen, N. Nguyen, M. Nodell, S. Pan, J. Peck, M. Peterson, W. Rowe, R. Sanders, J. Scott, M. Simpson, T. Smith, A. Sprague, T. Stockwell, R. Turner, E. Venter, M. Wang, M. Wen, D. Wu, M. Wu, A. Xia, A. Zandieh, X. Zhu, The sequence of the human genome. *Science* **291**, 1304–1351 (2001).
63. M. Srivastava, O. Simakov, J. Chapman, B. Fahey, M. E. A. Gauthier, T. Mitros, G. S. Richards, C. Conaco, M. Dacre, U. Hellsten, C. Larroux, N. H. Putnam, M. Stanke, M. Adamska, A. Darling, S. M. Degnan, T. H. Oakley, D. C. Plachetzki, Y. Zhai, M. Adamski, A. Calcino, S. F. Cummins, D. M. Goodstein, C. Harris, D. J. Jackson, S. P. Leys, S. Shu, B. J. Woodcroft, M. Vervoort, K. S. Kosik, G. Manning, B. M. Degnan, D. S. Rokhsar, The *Amphimedon queenslandica* genome and the evolution of animal complexity. *Nature* **466**, 720–726 (2010).
64. P. Martinez, S. C. Materna, J. Nam, P. Oliveri, F. Rizzo, J. Smith, D. Muzny, E. Sodergren, R. A. Gibbs, G. M. Weinstock, J. Fowler, R. Gill, C. Hamilton, J. Hernandez, S. Hines, J. Hume, A. Jolivet, C. Kovar, S. Lee, L. Lewis, G. Miner, M. Morgan, V. Lynne, Y. Shen, The genome of the sea urchin *Strongylocentrotus purpuratus*. *Science* **314**, 941–952 (2006).
65. J. A. Chapman, E. F. Kirkness, O. Simakov, S. E. Hampson, T. Mitros, T. Weinmaier, T. Rattei, P. G. Balasubramanian, J. Borman, D. Busam, K. Disbennett, C. Pfannkoch, N. Sumin, G. G. Sutton, L. D. Viswanathan, B. Walenz, D. M. Goodstein, U. Hellsten, T. Kawashima, S. E. Prochnik, N. H. Putnam, S. Shu, B. Blumberg, C. E. Dana, L. Gee, D. F. Kibler, L. Law, D. Lindgens, D. E. Martinez, J. Peng, P. A. Wigge, B. Bertulat, C. Guder, Y. Nakamura, S. Ozbek, H. Watanabe, K. Khalturin, G. Hemmrich, A. Franke, R. Augustin, S. Fraune, E. Hayakawa, S. Hayakawa, M. Hirose, J. S. Hwang, K. Ikeo, C. Nishimiya-Fujisawa, A. Ogura, T. Takahashi, P. R. H. Steinmetz, X. Zhang, R. Aufschnaiter, M. K. Eder, A. K. Gorny, W. Salvenmoser, A. M. Heimberg, B. M. Wheeler, K. J. Peterson, A. Böttger, P. Tischler, A. Wolf, T. Gojobori, K. A. Remington, R. L. Strausberg, J. C. Venter, U. Technau, B. Hobmayer, T. C. G. Bosch, T. W. Holstein, T. Fujisawa, H. R. Bode, C. N. David, D. S. Rokhsar, R. E. Steele, The dynamic genome of Hydra. *Nature* **464**, 592–596 (2010).
66. T. M. Williams, J. E. Selegue, T. Werner, N. Gompel, A. Kopp, S. B. Carroll, The regulation and evolution of a genetic switch controlling sexually dimorphic traits in *Drosophila*. *Cell* **134**, 610–623 (2008).
67. M. Rebeiz, J. W. Posakony, GenePalette: A universal software tool for genome sequence visualization and analysis. *Dev. Biol.* **271**, 431–438 (2004).
68. A. C. Groth, M. Fish, R. Nusse, M. P. Calos, Construction of transgenic *Drosophila* by using the site-specific integrase from phage phiC31. *Genetics* **166**, 1775–1782 (2004).
69. J. Bischof, R. K. Maeda, M. Hediger, F. Karch, K. Basler, An optimized transgenesis system for *Drosophila* using germ-line-specific phiC31 integrases. *Proc. Natl. Acad. Sci. U.S.A.* **104**, 3312–3317 (2007).
70. B. J. Vincent, G. R. Rice, G. M. Wong, W. J. Glassford, K. I. Downs, J. L. Shastay, K. Charles-obi, M. Natarajan, M. Gogol, J. Zeitlinger, M. Rebeiz, An atlas of transcription factors expressed in male pupal terminalia of *Drosophila melanogaster*. *G3* **9**, 3961–3972 (2019).
71. J. Schindelin, I. Arganda-carreras, E. Frise, V. Kaynig, T. Pietzsch, S. Preibisch, C. Rueden, S. Saalfeld, B. Schmid, J. Tinevez, D. J. White, V. Hartenstein, P. Tomancak, A. Cardona, Fiji—An open source platform for biological image analysis. *Nat. Methods* **9**, 676–682 (2019).

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A Notch signal required for a morphological novelty in *Drosophila* has antecedent functions in genital disc eversion

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